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Git Pull

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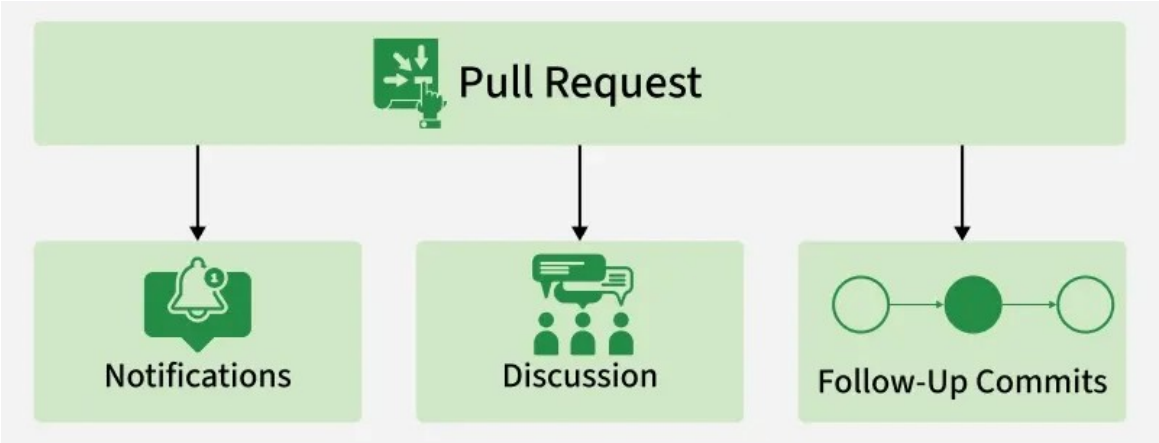
Git pull is a command which is used to fetch and integrate the changes which are present in the remote repository to the local repository.

- Git pull is a combination of `git fetch` and `git merge`.
- It is used to update the local branch with changes from the remote repository branch.
- Ensures that all developers have the latest code on their local machines.
- Allows developers to work with newly updated code without conflicts.

syntax of git pull command:

```
git pull [remote-name] [branch-name]
```



The image is a simple diagram explaining the components of a Pull Request.

- At the top, there is a “Pull Request” box, indicating the main action in Git/GitHub workflows.
- Three arrows point downward from the Pull Request, leading to

three components:

1. **Notifications** – Indicates that team members receive alerts when a pull request is created or updated.
2. **Discussion** – Represents collaboration where team members can comment, review code, and provide feedback.
3. **Follow-Up Commits** – Shows that additional commits can be made to the pull request based on feedback or changes before merging.

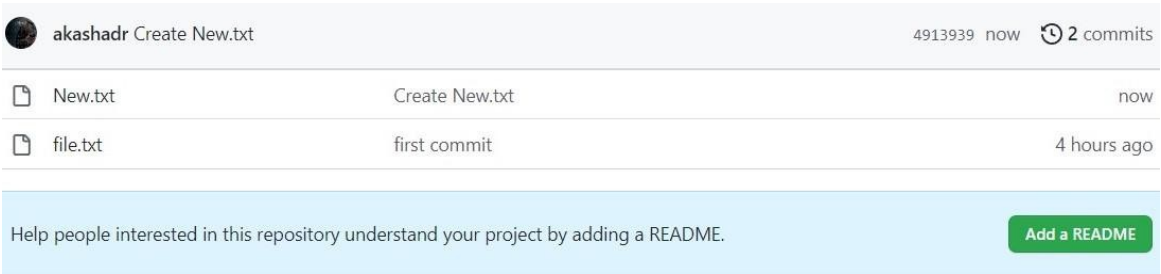
Overall, the diagram visually explains that a Pull Request involves alerts, communication, and iterative updates before the final merge.

How Git Pull Works?

Git Pull is a command used to update the local version of a repository from a remote repository. It is a mixture of two other commands:

1. git fetch
2. git merge

- **Stage 1:**`git pull` runs `git fetch` for the current branch (HEAD) to download changes.
- **Stage 2:** It performs a merge, creating a new merge commit and updating HEAD.
- **Example:** If you pushed `file.txt` and a teammate added `New.txt` on the remote, `git pull` fetches and merges `New.txt` into your local repository.



Now let's say you have created a new file "crio.txt" (crio means newborn) and after performing a commit when you try to push the

file into the remote repository, git will not allow you to push your code because the remote repository contains a task that you do not have locally on your computer.

```
Akash Jha@LAPTOP-LJJ1U61G MINGW64 ~/Desktop/Git (master)
$ git add "crio.txt"

Akash Jha@LAPTOP-LJJ1U61G MINGW64 ~/Desktop/Git (master)
$ git status
On branch master
Your branch is up to date with 'crio/master'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   crio.txt
```

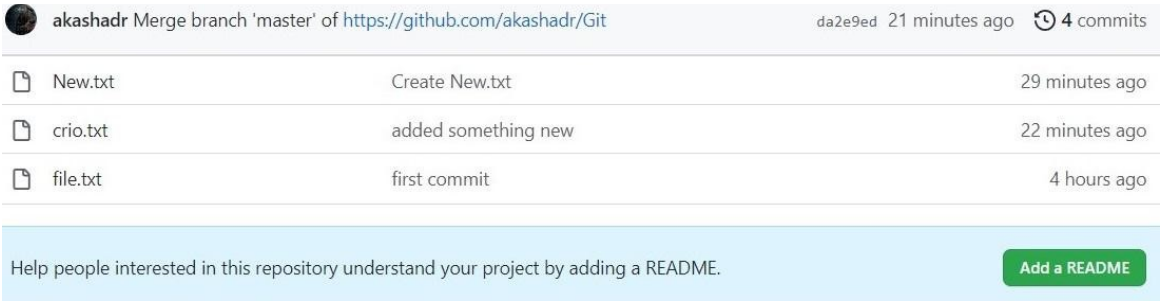
```
Akash Jha@LAPTOP-LJJ1U61G MINGW64 ~/Desktop/Git (master)
$ git commit -m "added something new"
[master 85007c7] added something new
 1 file changed, 1 insertion(+)
 create mode 100644 crio.txt

Akash Jha@LAPTOP-LJJ1U61G MINGW64 ~/Desktop/Git (master)
$ git push origin master
To https://github.com/akashadr/Git.git
 ! [rejected]        master -> master (fetch first)
error: failed to push some refs to 'https://github.com/akashadr/Git.git'
hint: Updates were rejected because the remote contains work that you do
hint: not have locally. This is usually caused by another repository pushing
hint: to the same ref. You may want to first integrate the remote changes
hint: (e.g., 'git pull ...') before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
```

In this case, `git pull` is used to fetch and download content from the remote repository to the local repository. After running `git pull origin master` where `origin` is the default remote and `master` is the branch your local repository is updated, allowing you to push your changes to the remote repository successfully.

```
Akash Jha@LAPTOP-LJJ1U61G MINGW64 ~/Desktop/Git (master)
$ git pull origin master
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
Unpacking objects: 100% (3/3), 655 bytes | 3.00 KiB/s, done.
From https://github.com/akashadr/Git
 * branch            master     -> FETCH_HEAD
   5a457ea..4913939  master     -> origin/master
Merge made by the 'ort' strategy.
 New.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 New.txt

Akash Jha@LAPTOP-LJJ1U61G MINGW64 ~/Desktop/Git (master)
$ git push origin master
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 4 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 615 bytes | 205.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/akashadr/Git.git
   4913939..da2e9ed  master -> master
```



Git Pull and Syncing

The main purpose of git pull command is to make the synchronization between the remote repository and the local repository. The term syncing refer to the updating the changes from remote repository.

Git pull command plays an major role in updating the local repository with the help of remote repository which will help in maintains the sync between the local and remote repository.

- **Git Fetch:** Fetches the latest changes form the remote repository to the local branch without integrating them.
- **Git merge:** It will integrates the coinages fetched from the remote repository.

Pulling via Rebase

If you want to maintain the your commits neat and linear order then you can use the Pulling via Rebase method follow the steps mentioned below.

Step 1: Lets consider your having the local commits with the names as

A---B---C

Step 2: Know fetch the changes that you want from the remote repository by using the following command. Fetch the changes form the remote origin.

```
git fetch main
```

Step 3: Pull changes via rebase by using the following command.

```
git pull --rebase origin main
```

Step 4: After completion of the git rebase.

Here the A--B--C are the commits which are available in the local repository and D---E---F are the changes form the remote repository. Know your local changes are on the top of the fetched changes.

A'--B'--C' (rebasing your changes)
**
D---E---F (remote branch)

Git Pull Examples

Example 1: To pull from the default remote and current branch

```
git pull
```

Example 2: Pulling From a Specific Remote and Branch

```
git pull <name of the branch>
```

Example 3: Pulling and Rebasing.

```
git pull --rebase
```

Git Pull on Remotes

Git pull remotes is an shorthand command of the git fetch and git merge or git rebase git pull will fetch the branch which is available in the remote repository and then it will integrate with the local repository the command which is used to git pull is as follows.

```
git pull
```

Git Pull Vs Git Fetech

Git Pull	Git Fetch
Git pull is the combination of two commands git fetch and git merge.	Git fetch command will fetch the remote repository to the local machine.
It will fetch the changes and merges with the repository.	It will only fetch the branch without any merge operation.
A more efficient method for updating your local branch is to use git pull.	Git fetch is frequently used as a first step before using 'git merge' to combine updates.

Suggested Quiz 5 Questions

What does the `git pull` command do?

A Uploads your commits to the remote repository

B Fetches and merges changes from the remote repository into your local branch

C Deletes outdated commits from the remote

D Only fetches changes without integrating them

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